CSE 3006: Computer Networks

Practical-07

Experiment: Ping-Traceroute-Telnet

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Semester: Winter 2025-26

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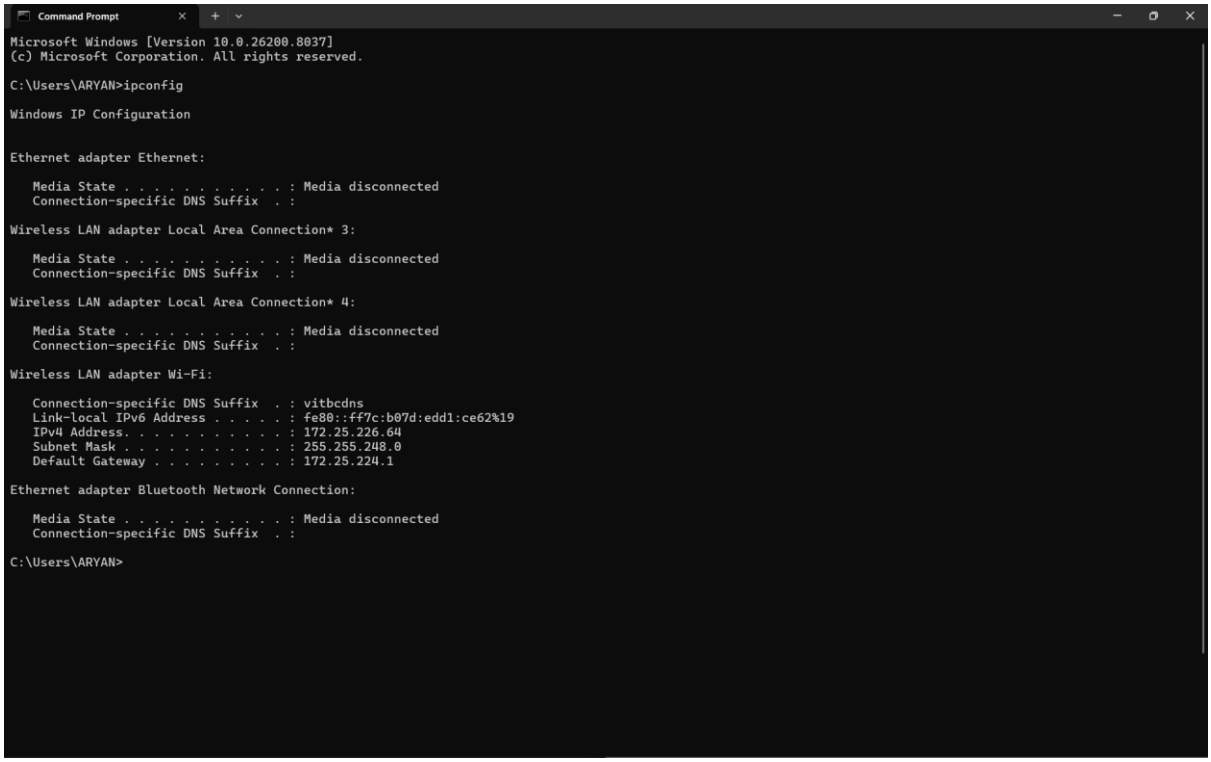
STEP 1: IP Check (ifconfig / ipconfig)



```
Command Prompt

Microsoft Windows [Version 10.0.26200.8037]
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C:\Users\ARYAN>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 3:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 4:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : vitbcdns
   Link-local IPv6 Address . . . . . : fe80::ff7c:b07d:edd1:ce62%19
   IPv4 Address. . . . . . . . . . . : 172.25.226.64
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 172.25.224.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\ARYAN>
```

STEP 2: Ping Google

```
C:\Users\ARYAN>ping www.google.com

Pinging www.google.com [142.251.210.132] with 32 bytes of data:
Reply from 142.251.210.132: bytes=32 time=232ms TTL=113
Reply from 142.251.210.132: bytes=32 time=233ms TTL=113
Reply from 142.251.210.132: bytes=32 time=233ms TTL=113
Reply from 142.251.210.132: bytes=32 time=233ms TTL=113

Ping statistics for 142.251.210.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 232ms, Maximum = 233ms, Average = 232ms
```

Ping uses ICMP protocol to test connectivity. All packets were successfully transmitted and received, indicating proper internet connectivity.

STEP 3: Traceroute

```
C:\Users\ARYAN>tracert www.google.com

Tracing route to www.google.com [142.251.210.132]
over a maximum of 30 hops:

  1     3 ms    <1 ms    15 ms  172.25.224.1
  2    <1 ms    <1 ms    <1 ms  172.25.90.2
  3     3 ms     3 ms     3 ms  139.167.143.181
  4     *        *        *     Request timed out.
  5    88 ms    21 ms    53 ms  172.31.2.71
  6    26 ms    22 ms    23 ms  173.194.121.8
  7    23 ms    23 ms    24 ms  192.178.84.191
  8    22 ms    24 ms    22 ms  142.250.209.70
  9    35 ms    22 ms     *     192.178.254.231
 10    75 ms    75 ms    75 ms  192.178.120.99
 11     *      101 ms     *     172.253.178.90
 12   148 ms   146 ms   146 ms  142.251.50.178
 13   233 ms   233 ms   234 ms  192.178.250.213
 14   239 ms   234 ms   234 ms  108.170.230.120
 15   239 ms   238 ms   238 ms  74.125.243.87
 16   234 ms   233 ms   233 ms  142.251.60.97
 17   232 ms   233 ms   232 ms  dclaxa-as-in-f4.1e100.net [142.251.210.132]

Trace complete.
```

Traceroute shows the path taken by packets from source to destination. Multiple hops indicate intermediate routers in the network.

STEP 4: Telnet

Telnet is a remote login protocol used to access another computer over a network. It works on port 23 and is part of the TCP/IP suite.

- Uses TCP protocol
- Not secure (no encryption)
- Replaced by SSH

Q1 (Analysis of commands)

Answer:

- The `ipconfig` command displays the IP configuration of the system including IP address, subnet mask, and default gateway.
- The `ping` command uses ICMP packets to test connectivity with a remote host (www.google.com). The successful replies indicate proper network connectivity.
- The `tracert` command shows the path taken by packets to reach the destination. It displays multiple hops representing intermediate routers.
- Telnet is a remote login protocol that allows users to access another system over a network using port 23.

Q2 (Observation)

The system successfully communicated with the remote server using ping. Traceroute showed multiple hops, indicating routing through intermediate devices. Telnet is used for remote access but is insecure.